

Multi-Agent Workflow Implementation

Summary

Demonstrate mastery of LangGraph by building a complete multi-agent system that solves a real-world problem through coordinated agent collaboration, persistent state management, and intelligent workflow orchestration.

Project Instructions

You will design and implement a production-ready multi-agent system that addresses a complex business problem requiring coordination between multiple specialized AI agents. Your system must demonstrate advanced LangGraph concepts including state management, conditional routing, error handling, and multi-agent coordination.

Select one of these business scenarios or propose your own:

→ Content Creation Pipeline: Coordinate agents for research, writing, editing, fact-checking, and publishing

→ Customer Support Automation: Orchestrate agents for inquiry classification, knowledge retrieval, response generation, and escalation management

→ Financial Analysis System: Coordinate agents for data collection, trend analysis, risk assessment, and report generation

→ Software Development Assistant: Orchestrate agents for code analysis, documentation, testing, and optimization recommendations

Technical Requirements

Multi-Agent Architecture (25 points)

- Implement at least 4 specialized agents with distinct responsibilities
- Design clear coordination patterns between agents
- Demonstrate agent collaboration that creates emergent intelligence
- Include at least one coordinator or orchestration component

Advanced State Management (25 points)

- Design comprehensive state schema that supports all workflow needs
- Implement persistent context that agents can build upon
- Include state evolution that improves system performance over time
- Provide state observability for debugging and monitoring

Conditional Logic and Error Handling (25 points)

- Implement sophisticated conditional routing based on multiple factors
- Include graceful degradation and fallback mechanisms
- Design retry logic and circuit breaker patterns
- Handle edge cases and unexpected inputs robustly

Production Readiness (25 points)

- Include comprehensive logging and monitoring capabilities
- Implement proper error handling and recovery mechanisms
- Design for scalability and maintainability
- Provide clear documentation and usage examples

Deliverables

System Implementation

- Complete LangGraph workflow implementation
- All agent implementations with clear interfaces
- State management and coordination logic
- Error handling and resilience mechanisms

Architecture Documentation

- System architecture diagram showing agent relationships
- State schema documentation with field descriptions
- Workflow logic explanation with decision trees
- API documentation for external integrations

Demonstration Materials

- Working demo with sample inputs and outputs
- Test cases showing different workflow paths
- Error handling demonstrations
- Performance and scalability analysis

Reflection Report

- Design decisions and trade-offs explanation
- Challenges encountered and solutions implemented
- Lessons learned about multi-agent coordination
- Future enhancement opportunities

Evaluation Criteria

Technical Excellence Quality of implementation and use of LangGraph features

Architectural Sophistication Thoughtful design of agent coordination and state management

Production Readiness Robustness, error handling, and operational considerations

Innovation Creative solutions and advanced coordination patterns

Documentation Quality Clear explanation of design decisions and system behavior

Submission Format

Submit a comprehensive package including:

- Source code with clear organization and comments
- README with setup and usage instructions
- Architecture documentation (PDF or Markdown)
- Demo video or interactive notebook
- Reflection report (1000-1500 words)

This capstone project demonstrates your mastery of LangGraph's core concepts and your ability to apply them to solve complex, real-world problems through sophisticated multi-agent coordination.